Band-1 Zauner SIC-POVM — One Dimension Probe (Collapse 2.31)

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Band-1: Zauner SIC-POVM in One Dimension — Honest Probe

Honesty stamp. This note does **not** claim that SIC-POVMs exist for all dimensions n. It does **not** claim a proof of Zauner's conjecture. It does **not** claim a Clay-millennial solution or a Theory of Everything. **GATE-04** is locked out of scope.

> **Collapse target (2.31):** *Zauner SIC-POVM in one new dimension — not all dimensions. One n with a certified fiducial.*
> **Box certified:** $d = 2$ (qubit tetrahedron SIC + Weyl–Heisenberg orbit; Welch bound saturated).
> **Zauner conjecture (all d): OPEN.**

Abstract

A SIC-POVM in dimension d is a minimal informationally complete POVM with d^2 rank-one effects whose projectors are equiangular: $|\langle\psi_i|\psi_j\rangle|^2 = 1/(d+1)$ for $i \neq j$. Zauner (1999) conjectured that a Weyl–Heisenberg (WH) covariant SIC exists for every $d \geq 2$. Existence for **all** d remains unproved. Exact fiducials are known for many finite d; numerical SICs are known much farther. Conditional constructive programs (Stark units / Appleby–Flammia–Kopp 2025) do **not** close the conjecture unconditionally.

This Band-1 probe (i) ledgers the literature status through 2024–2026, (ii) constructs and verifies a known exact SIC in **one** dimension ($d = 2$) on-box, (iii) checks the Welch bound / 2-design frame potential, and (iv) leaves the all-d question **OPEN**.

Problem (collapse 2.31 → Band-1)

1. Certify **one** dimension with an explicit fiducial / SIC orbit (Welch check).
2. Do **not** claim existence for all n.
3. Cite known exact / numerical status and 2025 conditional constructions honestly.
4. Keep Clay / ToE / GATE-04 off the claim surface.

Literature ledger (web check; not re-proved on box)

Item	Status	Notes
Zauner conjecture (WH SIC for every $d \geq 2$)	**OPEN**	Zauner dissertation 1999; Wikipedia *SIC-POVM* unsolved box
Exact fiducials	Known for all $d = 2, \ldots, 53$ and many larger (lit tally ~ 115 values of d, some as large as thousands)	Scott/Grassl / Appleby–Flammia–Waldron lineage; Flammia Exact SICs page
Numerical WH SICs	All d through ~ 193 and selected larger (historical Scott extensions to 151+; later catalogs farther)	Fuchs–Hoang–Stacey *Axioms* 2017 survey; subsequent extensions

Item	Status	Notes
Hoggar lines ($d = 8$ exceptional)	Known	Non-WH in the usual sense / exceptional SIC
Stark-unit algorithms (special $d = n^2 + 3$ families)	Computational constructions to very large special d (e.g. tens of thousands in published lines)	Appleby–Bengtsson–Grassl–Harrison–McConnell *J. Math. Phys.* 2022; 2024 follow-ons — **not** all-d proof
Appleby–Flammia–Kopp arXiv:2501.03970 (2025)	**CONDITIONAL** constructive approach	Ghost SIC + Stark + Twisted Convolution ⇒ Zauner; conditional, with numerical cross-checks (incl. new $d = 100$ examples) — **not** an unconditional proof
Existence for all d	**OPEN**	Do not elevate conditional / computational catalogs to "proved for all n"

Canonical portals. Wikipedia *SIC-POVM*; Fuchs–Hoang–Stacey, *The SIC Question: History and State of Play*, Axioms 6 (2017); Scott–Grassl catalogs; arXiv:2501.03970.

Definition and Welch bound (one page)

Unit vectors $\{|\psi_i\rangle\}_{i=1}^{d^2} \subset \mathbb{C}^d$ form a SIC when

$$|\langle\psi_i|\psi_j\rangle|^2 = \frac{1}{d+1} \qquad (i \neq j),$$

equivalently $F_i = |\psi_i\rangle\langle\psi_i|/d$ is a SIC-POVM ($\sum_i F_i = I$).

Welch bound. For N unit vectors in $\mathbb{C}^d$,

$$\frac{1}{N^2}\sum_{i,j} |\langle\psi_i|\psi_j\rangle|^2 \geq \frac{1}{d},$$

with equality for tight frames. A SIC saturates Welch with $N = d^2$ and constant off-diagonal overlap $1/(d+1)$. Equivalently the 2-design frame potential equals $2d^3/(d+1)$.

Box experiment

Lab: `experiments/band1/zauner_sic/`
Code: `verify_sic_d2.py` → `results.json`

Certified dimension: $d = 2$

1. **Tetrahedron SIC** (standard Bloch regular tetrahedron / Wikipedia vectors): four unit vectors in $\mathbb{C}^2$.
2. **WH orbit:** fiducial $|\phi\rangle$ with magnitudes matching the Renes–Appleby qubit form; orbit under Weyl–Heisenberg $W(p, q)$ for $(p, q) \in \mathbb{Z}_2 \times \mathbb{Z}_2$.

Check	Tetrahedron	WH orbit
$N = d^2 = 4$	yes	yes
$\max_i \big\lvert\lvert\langle\psi_i\rvert\psi_i\rangle\rvert^2 - 1\big\rvert$	$\sim 10^{-16}$	$\sim 10^{-16}$
$\max_{i\neq j} \big\lvert\lvert\langle\psi_i\rvert\psi_j\rangle\rvert^2 - 1/3\big\rvert$	$\sim 10^{-16}$	$\sim 10^{-16}$
$\lVert\sum_i F_i - I\rVert_\infty$	$\sim 10^{-16}$	$\sim 10^{-16}$

Check	Tetrahedron	WH orbit
Welch mean abs^2 $= 1/d = 1/2$	saturated	saturated
Frame potential $_4$ vs $2d^3/(d+1)$	match	match
is_SIC	**PASS**	**PASS**

Wall ≈ 0.002 s on box.

Secondary (not the collapse claim)

$d = 3$ Hesse-type fiducial $(0, 1, -1)/\sqrt{2}$ WH orbit: on-box **PASS** (overlap target 1/4). Recorded only as a literature cross-check; the **certified collapse dimension for this probe is** $d = 2$.

Verdict

Claim	Result
Certified SIC fiducial / orbit in **one** dimension on box?	**YES** — $d = 2$
Proved SIC for all n?	**NO**
Zauner conjecture status	**OPEN**
2025 Stark/ghost constructive program	**CONDITIONAL only** (not elevated)
Clay / ToE / GATE-04	**out of scope / LOCKED**

Next open (honest)

- **All-d Zauner conjecture:** still **OPEN**.
- Natural next Band-1 increment: certify one larger exact dimension **on this box** (e.g. $d = 4$ or $d = 5$ fiducial from published exact catalogs) — still **one** n, not all n.
- Do not treat arXiv:2501.03970 as closed Zauner; label **CONDITIONAL** until Stark + Twisted Convolution are settled (outside this probe).

References (secondary)

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3. C. A. Fuchs, M. C. Hoang, B. C. Stacey, *Axioms* 6 (2017) 21 — survey.
4. A. J. Scott, M. Grassl, *J. Math. Phys.* 51 (2010) 042203; Scott, arXiv:1703.03993.
5. M. Appleby, S. T. Flammia, G. S. Kopp, arXiv:2501.03970 (2025) — conditional construction.
6. Wikipedia, *SIC-POVM* (accessed 2026-09-11 house web check).

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